

John Fallin

470-292-6482 | jfallin3@gatech.edu | [linkedin.com/in/john-fallin-b51879274](https://www.linkedin.com/in/john-fallin-b51879274) | github.com/CaydenFallin | <https://kaderblast-portfolio.onrender.com/> |

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript (p5.js), Processing
Frameworks & Tools: Django, Unity, JAX, Git, VS Code
Creative Tools: DAW (Waveform), OBS, Davinci Resolve, Blender, Twine
Concepts: Game Design, 3D Modeling, AI/ML, Full-Stack Web Development, Video Production

EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computational Media, GPA: 3.5 Aug. 2024 – May 2028
• **Relevant Coursework:** Data Structures & Algorithms, Objects & Design, Intro to AI, Intro to Computational Media, Intro to Game Studies, Privacy & Technology Policy, Linear Algebra

PROJECTS

Sports RAG Chatbot | *Django, Python, RAG, AWS Bedrock* 04/2026
• Architected the full Django web framework for a team-built AI sports chatbot using a Retrieval-Augmented Generation pipeline to surface relevant documentation per query
• Implemented an integrated video player for AI-generated links and a developer-facing content management system

Competitive AI Game Agent | *Python, HMM, Minimax* 04/2026
• Co-developed a competitive two-agent AI for a tile-capture game using Hidden Markov Models for opponent tracking and predictive logic for strategic tile blocking
• Designed adversarial decision-making systems balancing offensive tile acquisition against defensive positioning

We Have 8 Minutes | *Unity, C#, Branching Narrative Design* 01/2026
• Designed and developed a narrative walking simulator with branching dialogue and a real-time countdown mechanic affecting story outcomes
• Wrote and performed the original soundtrack on guitar and vocals with three collaborating musicians, dynamically selected based on remaining in-game time
• Featured 2D hand-drawn sprite art integrated into a 3D environment; originally a course project expanded into a full public release

Tapedeck Head Renderer | *Python, OpenCV, MediaPipe, Pyrender* 05/2024
• Built a real-time AR face-tracking system that overlays a 3D Tapedeck model onto a face, with a multithreaded architecture separating pose estimation from rendering for live virtual camera and offline video processing.
• Diagnosed a rotation-matrix ambiguity across the pitch singularity causing tracking instability, identified the correct mode discriminator and implemented correction.

EXPERIENCE

Cashier & Front-of-House Lead 07/2021 – 09/2023
W&B Factory Acworth, GA
• Sole front-of-house employee responsible for all customer transactions and complaints without managerial support
• Managed cross-language communication with kitchen staff to coordinate orders and resolve service issues
• Handled opening and closing procedures independently across a 2+ year tenure

Martial Arts Instructor 03/2023 – 09/2023
Georgia Martial Arts and Self Defense Acworth, GA
• Designed and delivered structured curriculum to students across multiple skill levels
• Managed classroom environment and adapted teaching methods to individual learning styles
• Built consistent rapport with students and parents in a high-accountability setting

Line Cook 10/2023 – 07/2024
The Original Hot Chicken Woodstock, GA
• Executed food preparation under high-pressure, time-sensitive conditions in a team environment
• Maintained quality and consistency across high-volume service periods
• Led and organized team in timely cleaning and daily closure of business